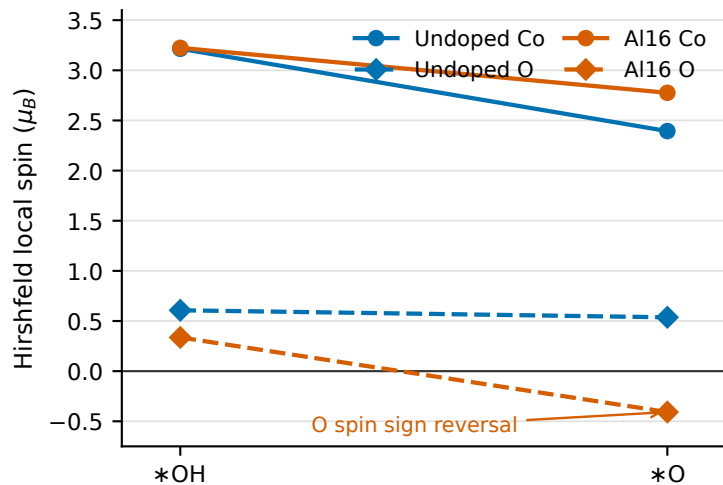
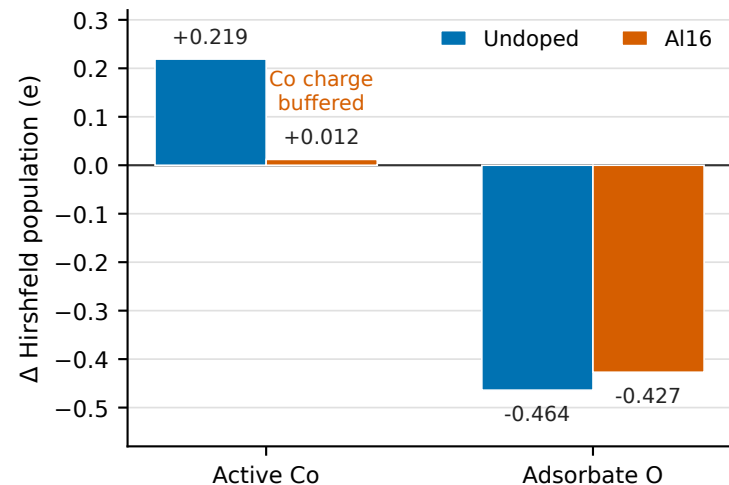


cDFT evidence: Al redirects the *OH → *O spin response

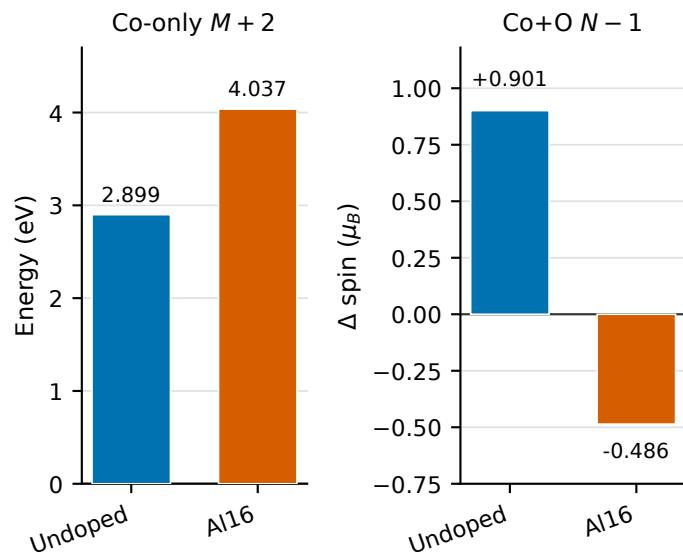
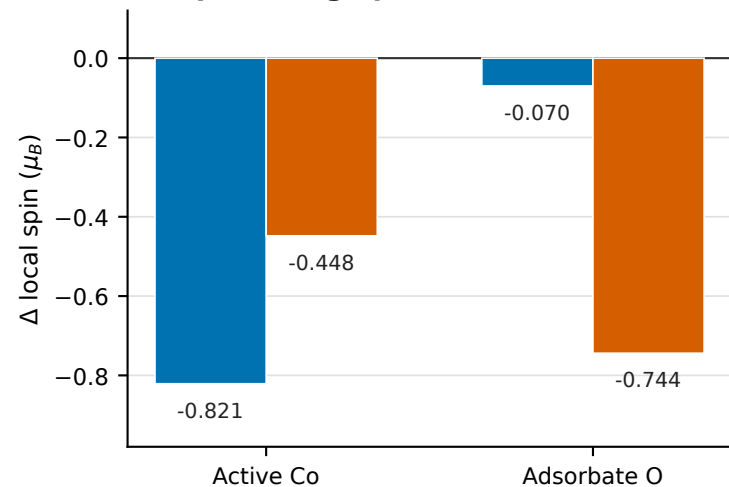
(a) Charge redistribution, *OH → *O

(b) Local-spin evolution



(c) Spin-change partition

(d) Constrained-state fingerprints for *O



Fixed-geometry comparison. Historical undoped Co32 and Al16 Co40 active sites are not geometrically identical.